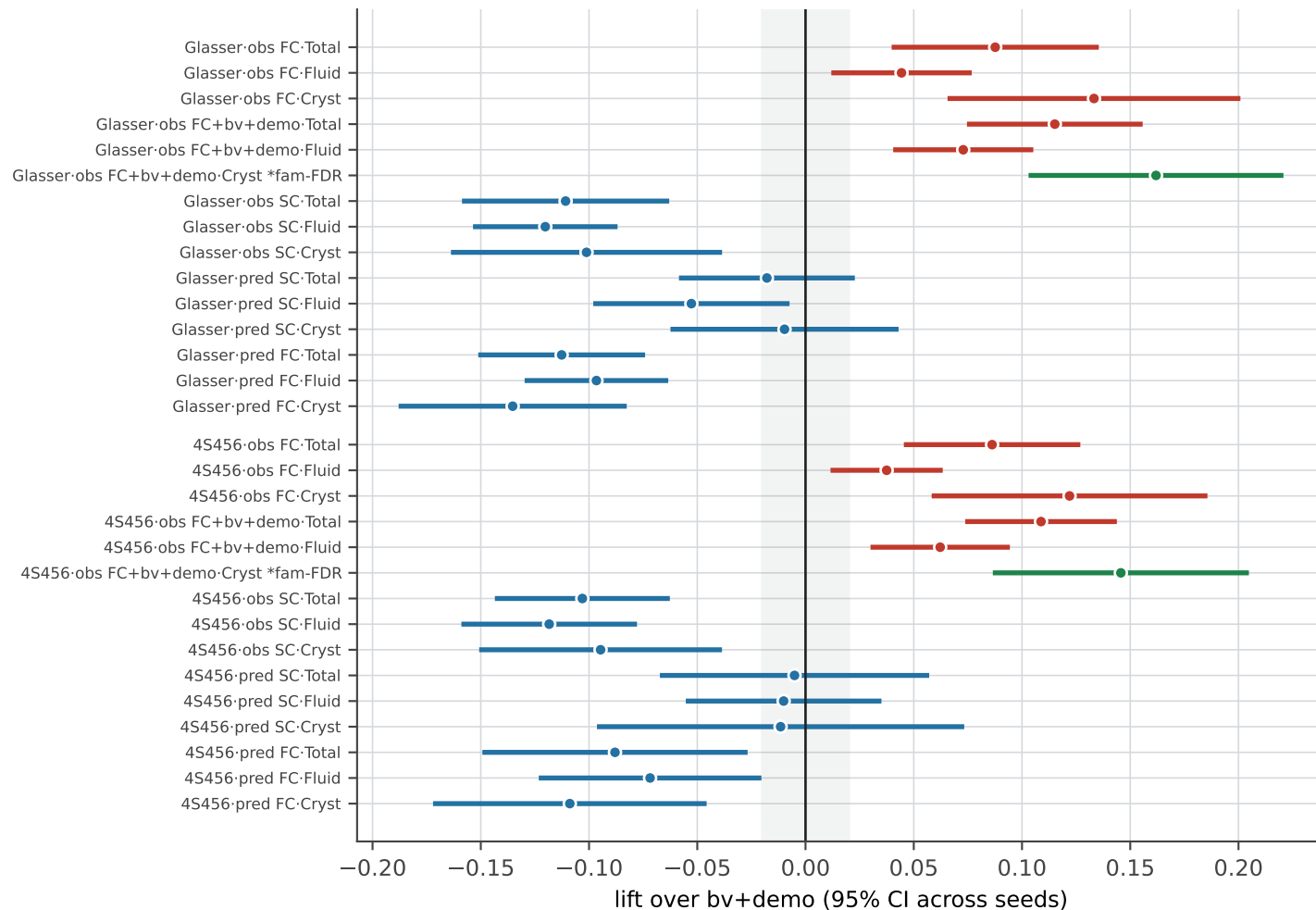


F11 · The observed-FC cognition lift is real but multiplicity-fragile (my robustness analysis)

A · downstream lifts with CIs; green = survives affirmative-family FDR



B

Multiplicity & power

Median MDE (80% power, n=10 seeds)

0.056 pearson

Positive lifts surviving whole-grid FDR

0 / 54

Positive lifts surviving family FDR

2 / 12

obs FC CogCryst — uncorrected p

≈0.03-0.04

obs FC CogCryst — whole-grid q

≈0.24 (fails)

obs FC+bv+demo CogCryst — family q

0.027/0.048 (passes)

*Design well-powered for ~0.1 lifts,
underpowered below ~0.03.*